

Over the years, my curiosity about Computational Fluid Dynamics has led me to spend a significant amount of time reading classical textbooks, reviewing research papers, and exploring existing CFD implementations. This curiosity largely comes from the way mathematics, physics, and scientific computing are combined to tackle complex fluid flow problems.

Through this process, I explored the fundamental differences between compressible and incompressible flow formulations, particularly how the governing Navier–Stokes equations change when density variations become significant. As a consequence, the numerical strategies used to solve these equations also differ, requiring distinct formulations, discretization approaches, and solution algorithms tailored to each regime.

I also examined several numerical discretization strategies used to solve these equations, including finite difference and finite volume methods, as well as different time-integration approaches such as explicit and implicit schemes. However, much of this understanding initially remained theoretical until I attempted to implement these methods myself. To truly internalize what I was learning, I decided to take the next step and develop my own solver.

While there is a large number of open-source codes available for compressible and turbulent flows, I found far fewer implementations of incompressible laminar solvers, particularly in generalized coordinates. This made the task more challenging, but also far more educational and rewarding.

This post marks the beginning of a series where I will document my journey, share my developments, and reflect on the numerical and physical insights behind them. I also hope it will open the door to valuable discussions and learning opportunities with others in the CFD community.